

Shibli Istiak Alam

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Research Interests

- Unsteady Aerodynamics
- Particle Image Velocimetry (PIV)
- Vortex Dynamics
- Low-Reynolds-Number Flows
- Bio-Inspired UAV Aerodynamics
- Gust Response

Research Experience

MS Thesis Research — Wake Dynamics of a NACA0012 Airfoil Undergoing a Single Oscillation at Low Reynolds Number.
Clarkson University | Aug 2023 – Present | Advisor: Dr. Douglas G. Bohl

- Conducted 2D PIV experiments on a NACA0012 airfoil undergoing single-cycle sinusoidal pitching at $Re_c \approx 5,200$, systematically varying Strouhal number and reduced frequency to isolate wake-formation mechanisms.
- Quantified vortex strength and structure (peak vorticity, circulation, core radius, convection speed) across 20+ kinematic cases, identifying pitch amplitude — not frequency — as the dominant driver of vortex strength.
- Developed a comparative framework linking isolated single-cycle wakes to continuous periodic pitching, demonstrating how wake-history effects progressively alter vortex evolution under repeated forcing.
- Operated and maintained a recirculating water tunnel and 2D PIV system, including laser alignment, camera calibration, seeding optimization, and troubleshooting synchronization/timing issues to ensure high-quality velocity field data.

Education

Clarkson University, *Master of Science in Mechanical Engineering*

Aug 2023 – Present

- GPA: 3.83/4.0
- Relevant Coursework: Advanced Fluid Mechanics, Special Topics in Fluid Mechanics, Advanced Turbulence, Computational Fluid Dynamics, Deep Learning.

Khulna University of Engineering & Technology, *Bachelor of Science in Mechanical Engineering*

Feb 2017 – May 2022

- Thesis: Numerical Study on Heat Transfer in an Enclosure Filled with a Nanofluid and Containing a Heat Source at the Center.

Publications

Manuscripts in Preparation

- Shibli Istiak Alam, D. G. Bohl, "Wake Dynamics of a NACA0012 Airfoil Undergoing a Single Oscillation at Low Reynolds Number" (*journal manuscript in preparation*)

Conference Publications

- SI Alam, SB Suja, MR Islam, "Analysis of Slot Jet impingement on a Moving Surface: A Numerical Approach", 7th International Conference on Mechanical, Industrial & Energy Engineering (ICMIEE-2022), Khulna, Bangladesh
- SI Alam, SB Suja, MS Islam, "Influence of Venturi on Heat Transfer Characteristics of Nanofluids Flowing Across Pipe", 6th International Conference on Mechanical Engineering and Renewable Energy (ICMERE-2021), Chattogram, Bangladesh
- SI Alam, MS Islam, MR Halder, "Heat Transfer in Square Enclosure Filled with SWCNT Nanofluid With a Square Heat Source at the center", 4th International Conference On Innovations In Mechanical Engineering (ICIME-2021), Hyderabad, India

Skills

Research and Analysis: LaVision DaVis (PIV), ImageJ (Pixel Analysis), DMC Motion Controller.

CAD and Animation Tools: Creo, SolidWorks, Siemens NX, AutoCAD, 3DS Max.

Simulation Tools: ANSYS, Abaqus, COMSOL.

Programming Language: MATLAB, Python.

Documentation and Presentation Tool: MS Office, LaTeX.

Work Experience

Graduate Teaching Assistant, *Clarkson University – Potsdam, NY*

Aug 2023 – Present

- Mentored undergraduate students in Fluid Mechanics, Measurement & Instrumentation, Experimental Methods, and Test Engineering (ES 330, ME/AE 201/301/401) through in-class lectures and lab demonstrations, and provided individualized guidance during office hours and written feedback on coursework.

Production Engineer, Bangladesh Steel Re-rolling Mills Ltd. – Chattogram, Bangladesh

- Designed and modified mechanical components (guides, vises) applying GD&T principles for precision fit and performance; contributed to a 4% reduction in operational delays through process and equipment optimization.

Mar 2023 – Jun 2023

Certifications

- Lean Six Sigma Green Belt
 - Fundamentals of Engineering (FE) – Mechanical, Passed
 - Certified SOLIDWORKS Professional (CSWP)
 - Certified SOLIDWORKS Associate (CSWA)
- May 2025

Sep 2024

Apr 2021

Feb 2021

Academic Projects

- Unsteady CFD Simulation of Flow Around a Square Cylinder at Re = 100.**

Jan 2026 - Jun 2026
- Developed a finite-volume CFD solver in MATLAB to simulate unsteady flow and vortex shedding around a square cylinder using pressure–velocity coupling and Rhie–Chow interpolation.
 - Evaluated drag and lift coefficients and compared predictions with commercial CFD results, analyzing the effects of mesh resolution and numerical schemes on solution accuracy.
- Deep Learning-Based Prediction of Li-Ion Battery Impedance Spectra.**

Jan 2026 - Jun 2026
- Developed deep neural network regression models using TensorFlow/Keras to predict real and imaginary components of Li-ion battery impedance from state-of-charge, frequency, and battery test conditions.
 - Achieved R² values of approximately 0.97 and 0.96 for real and imaginary impedance, respectively, demonstrating a data-driven surrogate approach for generating EIS responses and expanding battery diagnostic datasets.
- Design and Manufacturing of a Spinning W-Electrode for GTAW Process (S-TIG).**

Jan 2024 - Jun 2024
- Created detailed CAD models and assembly using SolidWorks of all the parts associated with the tungsten electrode.
 - Prepared project budget, bill of materials (BoM) and sourced materials for cost-effective production.
 - Coordinated with the machine shop for manufacturing and assembly, ensuring seamless communication among project members through regular updates and collaboration.
- Design & Animation of a Jet Engine.**

Jan 2024 - Jun 2024
- Made CAD models of different parts of a jet engine and assembled using Creo.
 - Rendered and created an animation with 3DS Max.
- Numerical Simulation of Turbulent Flow in a Channel.**

Jan 2024 - Jun 2024
- Simulated turbulent flow in rough channels using ANSYS Fluent, and validated predictions against experimental friction coefficients and boundary layer data.
- Finite Element Analysis of Airflow in a Constant Pressure Gradient Driven Duct.**

Aug 2023 - Dec 2023
- Developed a 2D FEM model applying Galerkin method to analyze velocity profiles in an air duct under no-slip conditions.
 - Performed error and convergence analysis using Richardson Extrapolation, comparing structured and unstructured mesh results for accuracy.
- Computational Analysis of Fluid Dynamics: Electronic Cooling, Boundary Layer Behavior & Similarity Solution to Falkner-Skan Equation.**

Aug 2023 - Dec 2023
- Analyzed boundary layer flow over a flat plate, electronic cooling in ducts using ANSYS Fluent.
 - Solved the Falkner-Skan equation using MATLAB to study velocity profiles and boundary layer thickness in wedge flow.

Standardized Test Results

GRE General Test Scores

Quantitative Reasoning	Verbal Reasoning	Analytical Writing
164	148	3.5

IELTS Academic Band Scores

Speaking	Listening	Reading	Writing	Overall
7.0	7.5	7.0	6.0	7.0